

Short statement of academic activity for **Dr. R. T. Mason**, 2005 September 1

Current Position: Associate Professor, Faculty of Education, University of Manitoba.

Principal Activities: I teach preservice, inservice, and graduate-level mathematics education, curriculum, and educational research courses. I am advisor for a number of graduate students. In the last year I served as Director of the Imperial Oil Academy for the Learning of Mathematics, Science and Technology; as Secretary and Program Coordinator for the Canadian Mathematics Education Study Group; as member of the editorial board for two journals (For the Learning of Mathematics, Journal of Educational Thought). I am an active researcher and scholar.

Research Grants received in the last two years:

2005-2010	Co-applicant, with others (G. Robinson, principal applicant). CRYSTAL initiative. National Sciences and Engineering Research Council.	\$1 000 000
2005-2008	Principal investigator, with J. McFeetors. Trajectories of students learning mathematics and science. Imperial Oil Academy, CRYSTAL.	\$53 300
2005-2006	Co-investigator with each of E. Janzen Roth, R. Hechter, C. Blagden, S. Erickson. Collaborative inquiries into the learning of mathematics and science. Imperial Oil Academy.	\$15847
2005-2008	Co-investigator, with A. Kajander, Lakehead U. Teachers' evolving mathematical understandings. CRYSTAL-UM.	\$59600
2004-2008	Principal investigator, with eight colleagues. Imperial Oil Academy for the Learning of Mathematics, Science, and Technology. Imperial Oil Foundation.	\$500 000
2004-2006	Principal investigator, with four colleagues. ArtsSmarts Manitoba Research Project. ArtsSmarts Secretariat, Manitoba Education, Imperial Oil Academy.	\$25 000

Publications, last three years, trimmed to fit space limitations but portray range.

Mason, R. T. (2005, in press). A kinder mathematics for Nunavut. In Yatta Kanu (Ed.), *Curriculum as Cultural Practice*. Toronto: OISE/UT Press.

McFeetors, P. J. & Mason, R. T. (2005, in press). Voice and success in non-academic mathematics courses. *For the Learning of Mathematics*, 25(3), 18-25.

Mason, R. T. (Ed.) (2005). Program, *Canadian Mathematics Education Study Group Conference, Ottawa, 2005*. Winnipeg: University of Manitoba.

Mason, R. T. (2005) Assessment for face validity, Carpenter Essential Skills Assessment. Winnipeg: Manitoba Advanced Education and Training Apprenticeship.

Mason, R. T. (2004). Will the real Isaac Newton please stand up? Toward an effective use of the historical icons of science in classrooms. In Don Metz (Ed.) *Seventh International History and Philosophy and Science Teaching Conference Proceedings* (pp. 580-595). Winnipeg, MB: University of Winnipeg.

Mason, R. T. & McFeetors, P. J. (2004). Voice and success in non-academic mathematics courses. In Douglas E. McDougall & John A. Ross (Eds.), *Proceedings of the twenty-sixth annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics* (pp. 745-756). Toronto: OISE/UT.

Mason, R. T. (2004). Dear Enrico: Why do we call them Fermi questions? *CMESG/GCEDM*

- Newsletter* 21(1), 5-7.
- Mason, R. T., Lewthwaite, B., & McMillan, B. (2004). The challenge of simplicity and the simplicity of challenge: A review of Amos and Boohan's Aspects of Teaching Secondary Science. *Interchange* 35(2), 269-272.
- Mason, R. T. (Ed.) (2003). Proceedings, *Fourth Annual Seminar in the History of Science in Education*. Winnipeg: University of Manitoba.
- Mason, R. T. (2003). Introducing performance tasks for mathematics. *The Math Journal*. 32(1), 17-19.
- Mason, R. T. et al. (2003). *Grade 8 Mathematics Performance Tasks with Rubrics*. Winnipeg: River East School Division.
- Mason, R. T. & McFeetors, P. J. (2002). Interactive writing in mathematics class—Getting started. *Mathematics Teacher*. 95(5), 532-536.
- Mason, R. T. (2002). Driver actions and traffic collisions. In Dawn White & Dexter Harvey (Eds.), *Road Safety Learning Resources for Schools Senior 2* (Chapter 2, pp. 34-52). Manitoba Education, Training and Youth & Manitoba Public Insurance.
- Mason, R. T. (2002). Mathematics--By invitation only. *Proceedings, 2000 Annual Meeting Canadian Mathematics Education Study Group*, Montreal.

Presentations, last year only.

- Mason, R. T. & Janzen Roth, E. (2005, July). Can students think like Archimedes? An instructional design experiment to incorporate history of science in student learning. International History and Philosophy of Science for Science Teaching. Leeds, England.
- Mason, R. T. (2005, May). Curriculum about and for moral justice: Examples from mathematics. The Canadian Association for Curriculum Studies Annual Pre-Conference. University of Western Ontario, London, ON.
- Mason, R. T., Gourdeau, F., Kieran, C., Higginson, W. & Zazkis, R. (2005, May) Debate: Compulsory Mathematics. Canadian Mathematics Education Study Group, Ottawa.
- Mason, R. T. & McFeetors, J. (2005, April). Students' conceptions of success in non-academic mathematics courses. American Educational Research Association Conference, Montreal.
- Mason, R. T. (2005, March). Action research comes of age. *Manitoba Education Research Network Forum on Strengthening Student Transitions*. Brandon, MB.
- Janzen Roth, E. & Mason, R. T. (2005, March) Thinking like Archimedes: An instructional design experiment. *Manitoba Education Research Network Forum on Strengthening Student Transitions*. Brandon, MB.
- Mason, R. T. (2004, December) *Adapting narrative inquiry methods to lighthouse research with teachers*. Manitoba Education Citizenship and Youth. Winnipeg.
- Mason, R. T. & McFeetors, P. J. (2004, October) Voice and success in non-academic mathematics courses. *Psychology of Mathematics Education—North America*. Toronto.
- Mason, R. T. (2004 September). *Proposal writing for practicing teachers: A workshop for potential researchers and collaborators*. University of Manitoba.
- Mason, R. T. & Janzen Roth, E. (2004, July). The insight of Archimedes: Scaffolding students' approximation of pi. *International Institute on Units of Historical Presentation in Science Teaching*. Deutsches Museum, Munich, Germany.
- Mason, R. T. (2004, July). Geometry for Nunavut: A postmodern approach to the cognition of Archimedes. *Fifth International Conference for History of Science in Science Education*. Keszthely, Hungary.